PG No. 47 $T_{\mathrm{h}}(c)$ $m\bar{3}$ [cubic] (internal axial dipole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(1,1)}[g](A_u)$

** symmetry

1

** expression

$$\frac{\sqrt{3}G_x x}{3}+\frac{\sqrt{3}G_y y}{3}+\frac{\sqrt{3}G_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_{1,1}^{(1,-1)}[g](T_g)$, $\vec{\mathbb{G}}_{1,2}^{(1,-1)}[g](T_g)$, $\vec{\mathbb{G}}_{1,3}^{(1,-1)}[g](T_g)$

** symmetry

x

y

z

** expression

G_x

G_y

G_z

$\vec{\mathbb{G}}_{1,1}^{(1,1)}[g](T_g)$, $\vec{\mathbb{G}}_{1,2}^{(1,1)}[g](T_g)$, $\vec{\mathbb{G}}_{1,3}^{(1,1)}[g](T_g)$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{10}G_x\left(2x^2-y^2-z^2\right)}{10}+\frac{3\sqrt{10}G_y xy}{10}+\frac{3\sqrt{10}G_z xz}{10}$$

$$\frac{3\sqrt{10}G_x xy}{10}-\frac{\sqrt{10}G_y\left(x^2-2y^2+z^2\right)}{10}+\frac{3\sqrt{10}G_z yz}{10}$$

$$\frac{3\sqrt{10}G_x xz}{10}+\frac{3\sqrt{10}G_y yz}{10}-\frac{\sqrt{10}G_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{Q}}_{1,1}^{(1,0)}[g](T_u)$, $\vec{\mathbb{Q}}_{1,2}^{(1,0)}[g](T_u)$, $\vec{\mathbb{Q}}_{1,3}^{(1,0)}[g](T_u)$

** symmetry

x

y

z

** expression

$$\frac{\sqrt{2}G_y z}{2}-\frac{\sqrt{2}G_z y}{2}$$

$$-\frac{\sqrt{2}G_x z}{2}+\frac{\sqrt{2}G_z x}{2}$$

$$\frac{\sqrt{2}G_x y}{2}-\frac{\sqrt{2}G_y x}{2}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(1,0)}[g](E_g^{(a)})$

** symmetry

$$-\frac{\sqrt{2}x^2}{4}+\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}-\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{G_x yz\left(\sqrt{3}+i\right)}{2}-\frac{G_y xz\left(\sqrt{3}-i\right)}{2}-iG_z xy$$

$\vec{\mathbb{Q}}_2^{(1,0)}[g](E_g^{(b)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{G_x yz\left(\sqrt{3}-i\right)}{2}-\frac{G_y xz\left(\sqrt{3}+i\right)}{2}+iG_z xy$$

$\vec{\mathbb{Q}}_{2,1}^{(1,0)}[g](T_g)$, $\vec{\mathbb{Q}}_{2,2}^{(1,0)}[g](T_g)$, $\vec{\mathbb{Q}}_{2,3}^{(1,0)}[g](T_g)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}G_x\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{2}G_y xy}{2}+\frac{\sqrt{2}G_z xz}{2}$$

$$\frac{\sqrt{2}G_x xy}{2}-\frac{\sqrt{2}G_y\left(x-z\right)\left(x+z\right)}{2}-\frac{\sqrt{2}G_z yz}{2}$$

$$-\frac{\sqrt{2}G_x xz}{2}+\frac{\sqrt{2}G_y yz}{2}+\frac{\sqrt{2}G_z\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](E_u^{(a)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}+\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}-\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{G_x x\left(\sqrt{3}-3i\right)}{6}-\frac{G_y y\left(\sqrt{3}+3i\right)}{6}+\frac{\sqrt{3}G_z z}{3}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](E_u^{(a)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}+\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}-\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$\frac{iG_x x\left(3\sqrt{14}x^2+\sqrt{42}ix^2-7\sqrt{14}y^2+\sqrt{42}iy^2-2\sqrt{14}z^2-4\sqrt{42}iz^2\right)}{28}$$
$$+\frac{iG_y y\left(7\sqrt{14}x^2+\sqrt{42}ix^2-3\sqrt{14}y^2+\sqrt{42}iy^2+2\sqrt{14}z^2-4\sqrt{42}iz^2\right)}{28}+\frac{iG_z z\left(5\sqrt{14}x^2+3\sqrt{42}ix^2-5\sqrt{14}y^2+3\sqrt{42}iy^2-2\sqrt{42}iz^2\right)}{28}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](E_u^{(b)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{G_x x\left(\sqrt{3}+3i\right)}{6}-\frac{G_y y\left(\sqrt{3}-3i\right)}{6}+\frac{\sqrt{3}G_z z}{3}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](E_u^{(b)})$
** symmetry

$$-\frac{\sqrt{2}x^2}{4}-\frac{\sqrt{6}ix^2}{4}-\frac{\sqrt{2}y^2}{4}+\frac{\sqrt{6}iy^2}{4}+\frac{\sqrt{2}z^2}{2}$$

** expression

$$-\frac{iG_x x\left(3\sqrt{14}x^2-\sqrt{42}ix^2-7\sqrt{14}y^2-\sqrt{42}iy^2-2\sqrt{14}z^2+4\sqrt{42}iz^2\right)}{28}$$

$$-\frac{iG_y y\left(7\sqrt{14}x^2-\sqrt{42}ix^2-3\sqrt{14}y^2-\sqrt{42}iy^2+2\sqrt{14}z^2+4\sqrt{42}iz^2\right)}{28}-\frac{iG_z z\left(5\sqrt{14}x^2-3\sqrt{42}ix^2-5\sqrt{14}y^2-3\sqrt{42}iy^2+2\sqrt{42}iz^2\right)}{28}$$

$\vec{\mathbb{G}}_{2,1}^{(1,-1)}[g](T_u)$, $\vec{\mathbb{G}}_{2,2}^{(1,-1)}[g](T_u)$, $\vec{\mathbb{G}}_{2,3}^{(1,-1)}[g](T_u)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{2}G_y z}{2}+\frac{\sqrt{2}G_z y}{2}$$

$$\frac{\sqrt{2}G_x z}{2}+\frac{\sqrt{2}G_z x}{2}$$

$$\frac{\sqrt{2}G_x y}{2}+\frac{\sqrt{2}G_y x}{2}$$

$\vec{\mathbb{G}}_{2,1}^{(1,1)}[g](T_u)$, $\vec{\mathbb{G}}_{2,2}^{(1,1)}[g](T_u)$, $\vec{\mathbb{G}}_{2,3}^{(1,1)}[g](T_u)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

$$\sqrt{3}xy$$

** expression

$$\frac{5\sqrt{7}G_x xyz}{7}-\frac{\sqrt{7}G_y z\left(x^2-4y^2+z^2\right)}{7}-\frac{\sqrt{7}G_z y\left(x^2+y^2-4z^2\right)}{7}$$

$$\frac{\sqrt{7}G_x z\left(4x^2-y^2-z^2\right)}{7}+\frac{5\sqrt{7}G_y xyz}{7}-\frac{\sqrt{7}G_z x\left(x^2+y^2-4z^2\right)}{7}$$

$$\frac{\sqrt{7}G_x y\left(4x^2-y^2-z^2\right)}{7}-\frac{\sqrt{7}G_y x\left(x^2-4y^2+z^2\right)}{7}+\frac{5\sqrt{7}G_z xyz}{7}$$

* Harmonics for rank 3
$\vec{\mathbb{G}}_{3}^{(1,-1)}[g](A_g)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$G_x yz+G_y xz+G_z xy$$

$\vec{\mathbb{G}}_{3}^{(1,1)}[g](A_g)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{15}G_x yz\left(6x^2-y^2-z^2\right)}{6}-\frac{\sqrt{15}G_y xz\left(x^2-6y^2+z^2\right)}{6}-\frac{\sqrt{15}G_z xy\left(x^2+y^2-6z^2\right)}{6}$$

$\vec{\mathbb{G}}_{3,1}^{(1,-1)}[g](T_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(1,-1)}[g](T_g,1)$, $\vec{\mathbb{G}}_{3,3}^{(1,-1)}[g](T_g,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{\sqrt{15}G_x\left(2x^2 - y^2 - z^2\right)}{10} - \frac{\sqrt{15}G_yxy}{5} - \frac{\sqrt{15}G_zxz}{5}$$

$$-\frac{\sqrt{15}G_xxy}{5} - \frac{\sqrt{15}G_y\left(x^2 - 2y^2 + z^2\right)}{10} - \frac{\sqrt{15}G_zyz}{5}$$

$$-\frac{\sqrt{15}G_xxz}{5} - \frac{\sqrt{15}G_yyz}{5} - \frac{\sqrt{15}G_z\left(x^2 + y^2 - 2z^2\right)}{10}$$

$\vec{\mathbb{G}}_{3,1}^{(1,-1)}[g](T_g, 2)$, $\vec{\mathbb{G}}_{3,2}^{(1,-1)}[g](T_g, 2)$, $\vec{\mathbb{G}}_{3,3}^{(1,-1)}[g](T_g, 2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{G_x\left(y - z\right)\left(y + z\right)}{2} + G_yxy - G_zxz$$

$$-G_xxy - \frac{G_y\left(x - z\right)\left(x + z\right)}{2} + G_zyz$$

$$G_xxz - G_yyz + \frac{G_z\left(x - y\right)\left(x + y\right)}{2}$$

$\vec{\mathbb{G}}_{3,1}^{(1,1)}[g](T_g, 1)$, $\vec{\mathbb{G}}_{3,2}^{(1,1)}[g](T_g, 1)$, $\vec{\mathbb{G}}_{3,3}^{(1,1)}[g](T_g, 1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{G_x\left(8x^4 - 24x^2y^2 - 24x^2z^2 + 3y^4 + 6y^2z^2 + 3z^4\right)}{12} + \frac{5G_yxy\left(4x^2 - 3y^2 - 3z^2\right)}{12} + \frac{5G_zxz\left(4x^2 - 3y^2 - 3z^2\right)}{12}$$

$$-\frac{5G_xxy\left(3x^2 - 4y^2 + 3z^2\right)}{12} + \frac{G_y\left(3x^4 - 24x^2y^2 + 6x^2z^2 + 8y^4 - 24y^2z^2 + 3z^4\right)}{12} - \frac{5G_zyz\left(3x^2 - 4y^2 + 3z^2\right)}{12}$$

$$-\frac{5G_xxz\left(3x^2 + 3y^2 - 4z^2\right)}{12} - \frac{5G_yyz\left(3x^2 + 3y^2 - 4z^2\right)}{12} + \frac{G_z\left(3x^4 + 6x^2y^2 - 24x^2z^2 + 3y^4 - 24y^2z^2 + 8z^4\right)}{12}$$

$\vec{\mathbb{G}}_{3,1}^{(1,1)}[g](T_g, 2)$, $\vec{\mathbb{G}}_{3,2}^{(1,1)}[g](T_g, 2)$, $\vec{\mathbb{G}}_{3,3}^{(1,1)}[g](T_g, 2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{15}G_x\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{12}-\frac{\sqrt{15}G_yxy\left(2x^2-5y^2+9z^2\right)}{12}+\frac{\sqrt{15}G_zxz\left(2x^2+9y^2-5z^2\right)}{12}$$

$$-\frac{\sqrt{15}G_xxy\left(5x^2-2y^2-9z^2\right)}{12}+\frac{\sqrt{15}G_y\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{12}-\frac{\sqrt{15}G_zyz\left(9x^2+2y^2-5z^2\right)}{12}$$

$$\frac{\sqrt{15}G_xxz\left(5x^2-9y^2-2z^2\right)}{12}+\frac{\sqrt{15}G_yyz\left(9x^2-5y^2+2z^2\right)}{12}-\frac{\sqrt{15}G_z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](A_u)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{5}G_xx\left(y-z\right)\left(y+z\right)}{2}-\frac{\sqrt{5}G_yy\left(x-z\right)\left(x+z\right)}{2}+\frac{\sqrt{5}G_zz\left(x-y\right)\left(x+y\right)}{2}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,0)}[g](T_u,1)$, $\vec{\mathbb{Q}}_{3,2}^{(1,0)}[g](T_u,1)$, $\vec{\mathbb{Q}}_{3,3}^{(1,0)}[g](T_u,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{3}G_yz\left(4x^2-y^2-z^2\right)}{4}-\frac{\sqrt{3}G_zy\left(4x^2-y^2-z^2\right)}{4}$$

$$\frac{\sqrt{3}G_xz\left(x^2-4y^2+z^2\right)}{4}-\frac{\sqrt{3}G_zx\left(x^2-4y^2+z^2\right)}{4}$$

$$-\frac{\sqrt{3}G_xy\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{3}G_yx\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}_{3,1}^{(1,0)}[g](T_u,2)$, $\vec{\mathbb{Q}}_{3,2}^{(1,0)}[g](T_u,2)$, $\vec{\mathbb{Q}}_{3,3}^{(1,0)}[g](T_u,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\sqrt{5}G_xxyz+\frac{\sqrt{5}G_yz\left(2x^2+y^2-z^2\right)}{4}+\frac{\sqrt{5}G_zy\left(2x^2-y^2+z^2\right)}{4}$$

$$\frac{\sqrt{5}G_xz\left(x^2+2y^2-z^2\right)}{4}-\sqrt{5}G_yxyz-\frac{\sqrt{5}G_zx\left(x^2-2y^2-z^2\right)}{4}$$

$$\frac{\sqrt{5}G_xy\left(x^2-y^2+2z^2\right)}{4}-\frac{\sqrt{5}G_yx\left(x^2-y^2-2z^2\right)}{4}-\sqrt{5}G_zxyz$$